

Conversational AI Assignment 2 Report

Assignment: Conversational AI Assignment 2 - Parameter-Efficient Fine-Tuning

Domain: Banking support

Dataset: Merged banking instruction dataset with 480 samples, cleaned to 347

Split: Train 277 / Validation 35 / Test 35

Benchmark: 8 frozen prompts covering factual, formatting, reasoning, and out-of-scope cases

Technique: LoRA-based parameter-efficient fine-tuning with google/flan-t5-small

Artifacts generated:

- output/cleaned_dataset.json
- output/train.json
- output/validation.json
- output/test.json
- output/benchmark_prompts.json
- output/benchmark_results.json
- output/assignment_summary.json
- output/assignment_report_summary.json

Workflow summary:

1. Merge the banking instruction JSON files into a single flat dataset.
2. Clean duplicates and empty/truncated responses.
3. Create train/validation/test splits.
4. Freeze an 8-prompt benchmark for evaluation.
5. Prepare a LoRA-based training and evaluation pipeline for the SLM.

This report was generated from the scripts in the Banking-Data folder.